WATER PUMP

3. Coat shaft and insert with Lubriplate or equivalent. Carefully position seal assembly over bearing shaft and slinger. Install spring over seal, with large end of spring toward impeller.

4. With pump set in press so it is supported on the outer end of pump shaft, press impeller onto pump shaft.

CAUTION: Be sure assembly is supported properly to prevent damage to bearing and staking.

5. Press impeller onto shaft until outer face of impeller hub is .052"-.072" below face of pump body (fig. 2). Impeller must be positioned on shaft so pin hole in shaft is located midway between blades of impeller. Rotate impeller to be sure clearance exists between face of impeller blades and pump body.

6. Using the hole in the pump shaft as a jig, drill a .184" diameter hole through impeller hub. Hole should be located approximately midway between impeller blades. Drive pin into hole through shaft and impeller.

7. With pump positioned in press so it is supported on the impeller end of pump shaft, drive coupling and slinger assembly on opposite end of shaft until coupling is flush with end of shaft. Coupling must be a tight press fit on shaft since no drive key is used.

8. Position gasket to pump body at bolting flange for combination inlet and cover. Install cover over studs and up against body, with elbow pointing down, and pump outlet toward cylinder block. Tighten cover in place with nuts and lock washers. Test bind between impeller and cover by turning pump shaft by hand.

9. If pump drive coupling and slinger assembly was removed from the blower shaft, insert splined end of coupling into mating splines of blower shaft. Insert the screw tightly to hold coupling. Be sure coupling is fully in place before tightening screw; otherwise a preloading of the water pump bearing may cause damage.

INSTALLATION

1. Place water pump outlet packing flange over pump outlet, with flat machined face away from pump body. Place packing over outlet next to flange.

2. With inlet elbow opening pointing down, install hose clamp and water pump inlet seal over inlet elbow.

3. Position pump assembly at end of blower cover so that lugs register on the two drive couplings. In this position pump will be correctly positioned with respect to cylinder block and oil cooler. Secure with cap screws and lock washers.

4. Install water outlet packing and packing flange against cylinder block; then secure with cap screws and lock washers.

5. Slide pump inlet seal and hose clamp in place at junction of oil cooler and pump inlet elbow; then tighten clamp screws.

6. If water by-pass tube is attached to inlet it must be reconnected using new gasket and cap screws.

7. Fill cooling system and check for water leaks.

HYDRAULIC DRIVEN FAN

DESCRIPTION

A thermostatically-controlled hydraulic-driven fan that operates only when engine cooling is needed is used as standard on truck engines and as optional on coach engines. Two types of mounting are used—one at balance weight cover (fig. 7) and the other at front of crankshaft (fig. 8). The arrangement shown in figure 7 is driven by conventional belts and pulleys, while type shown in figure 8 is driven directly from crankshaft.

The fan drive consists of a driving and a driven torus member enclosed within an aluminum housing. Speed of fan is variable, depending upon operating temperature. When engine is cool, little or no power is transmitted to the driven torus and fan idles or may stop. When engine is hot, full power is transmitted to the fan to assist in cooling the engine.

OPERATION

A thermostatically-operated valve is installed in water return between radiator and oil cooler (fig. 9). The amount of oil retained in the torus housing is controlled by the valve, which is actuated by water temperature.

Under cold conditions, the valve is open permitting oil to return to the crankcase and no power is transmitted to fan by torus driven member. As the temperature of the coolant reaches a predetermined setting the valve starts to close, retaining oil in torus housing and power is gradually transmitted to fan.

The fan speed varies with the amount of oil retained in the coupling as determined by the temperature of water. The fan speed is varied gradually as temperatures change due to operating conditions.